QUESTION-11

Queue Using Array

main.c

Run

```c
#include <stdio.h>
#define MAX 5
int queue[MAX];
int front = -1, rear = -1;
void enqueue(int value) {
    if (rear == MAX - 1) {
        printf("Queue Overflow\n");
        return;
    }
    if (front == -1)
        front = 0;
    rear++;
    queue[rear] = value;
    printf("%d inserted into queue\n", value);
}
void dequeue() {
    if (front == -1 || front > rear) {
        printf("Queue Underflow\n");
        return;
    }
    printf("Deleted Element: %d\n",
        queue[front]);
    front++;
    if (front > rear)
```

Output

Clear

```
10 inserted into queue
20 inserted into queue
30 inserted into queue
Queue Elements: 10 20 30
Front Element: 10
Deleted Element: 10
Queue Elements: 20 30


=== Code Execution Successful ===
```

main.c

Run

```c
        front = rear = -1;
}
void display() {
    int i;
    if (front == -1) {
        printf("Queue is Empty\n");
        return;
    }
    printf("Queue Elements: ");
    for (i = front; i <= rear; i++) {
        printf("%d ", queue[i]);
    }
    printf("\n");
}
void peek() {
    if (front == -1)
        printf("Queue is Empty\n");
    else
        printf("Front Element: %d\n",
            queue[front]);
}
int main() {
    enqueue(10);
    enqueue(20);
```

Output

Clear


```
10 inserted into queue
20 inserted into queue
30 inserted into queue
Queue Elements: 10 20 30
Front Element: 10
Deleted Element: 10
Queue Elements: 20 30


=== Code Execution Successful ===
```


main.c

Run


```c
    }
    printf("Queue Elements: ");
    for (i = front; i <= rear; i++) {
        printf("%d ", queue[i]);
    }
    printf("\n");
}
void peek() {
    if (front == -1)
        printf("Queue is Empty\n");
    else
        printf("Front Element: %d\n",
            queue[front]);
}
int main() {
    enqueue(10);
    enqueue(20);
```


```c
    enqueue(30);
    display();
    peek();
    dequeue();
    display();
    return 0;
}
```

Output

Clear


```
10 inserted into queue
20 inserted into queue
30 inserted into queue
Queue Elements: 10 20 30
Front Element: 10
Deleted Element: 10
Queue Elements: 20 30


=== Code Execution Successful ===
```